TOR INDEPENDENCE PRESERVES TOR AMPLITUDE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof computes the derived pullback on stalks, applies the tor-independence base change comparison, derives part (1) from the pull-tor-amplitude lemma, and reads part (2) off part (1) via the tor amplitude characterization of flatness

Editorial note. The label lemma-tor-independence-and-tor-amplitude names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- tor amplitude of complexes computed on stalks (Cohomology on Sites, Lemma sites-cohomology-lemma-tor-amplitude-stalk) with the stalk theorem for exactness (Properties of Spaces, Theorem spaces-properties-theorem-exactness-stalks)
- the stalk formula for derived pullback along the base changed morphism
- the base change comparison of derived tensor products under tor independence (More on Algebra, Lemma more-algebra-lemma-base-change-comparison)
- the behaviour of tor amplitude under field/tensor base change (More on Algebra, Lemma more-algebra-lemma-pull-tor-amplitude)

Source reference. Original lemma 0661: sources/stacks-chow/tags.txt, source lines 7920–7920. Prerequisite pointer invoked at proof line(s) 5269; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 066L: sources/stacks-chow/tags.txt, source lines 7940–7940. Prerequisite pointer invoked at proof line(s) 5276; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DJJ: sources/stacks-chow/tags.txt, source lines 16979–16979. Prerequisite pointer invoked at proof line(s) 5255; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 04K5: sources/stacks-chow/tags.txt, source lines 5961–5961. Prerequisite pointer invoked at proof line(s) 5257; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Consider a cartesian square of algebraic spaces*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

over S. Assume g and f Tor independent.

1. *If $E \in D(\mathcal{O}_X)$ has tor amplitude in $[a, b]$ as a complex of $f^{-1}\mathcal{O}_Y$-modules, then $L(g')^*E$ has tor amplitude in $[a, b]$ as a complex of $f^{-1}\mathcal{O}_{Y'}$-modules.*
2. *If $\mathcal{G}$ is an $\mathcal{O}_X$-module flat over Y, then $L(g')^*\mathcal{G} = (g')^*\mathcal{G}$.*

Proof. We can compute tor dimension at stalks, see Cohomology on Sites, Lemma 0DJJ and Properties of Spaces, Theorem 04K5. If $\overline{x}'$ is a geometric point of X' with image $\overline{x}$ in X, then

$$(L(g')^*E)_{\overline{x}'} = E_{\overline{x}} \otimes^{\mathbf{L}}_{\mathcal{O}_{X,\overline{x}}} \mathcal{O}_{X',\overline{x}'}$$

Let $\overline{y}'$ in Y' and $\overline{y}$ in Y be the image of $\overline{x}'$ and $\overline{x}$. Since X and Y' are tor independent over Y, we can apply More on Algebra, Lemma 0661 to see that the right hand side of the displayed formula is equal to $E_{\overline{x}} \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,\overline{y}}} \mathcal{O}_{Y',\overline{y}'}$ in $D(\mathcal{O}_{Y',\overline{y}'})$. Thus (1) follows from More on Algebra, Lemma 066L. To see (2) observe that flatness of $\mathcal{G}$ is equivalent to the condition that $\mathcal{G}[0]$ has tor amplitude in $[0, 0]$. Applying (1) we conclude. □